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Semiconductor device and electronic system

Abstract

A semiconductor device includes a lower structure including a peripheral circuit, a lower insulating structure covering the peripheral circuit, and a pattern structure on the lower insulating structure; a stack structure including interlayer insulating layers and horizontal layers alternately stacked on the lower structure, wherein the horizontal layers include gate horizontal layers in a gate region of the stack structure and first insulating horizontal layers in a first insulating region of the stack structure; a memory vertical structure including a portion penetrating the gate horizontal layers; dummy vertical structures including a portion penetrating the gate horizontal layers; a first peripheral contact plug including a portion penetrating the first insulating region; and gate contact plugs on gate pads of the gate horizontal layers, wherein upper surface of the gate contact plugs and the first peripheral contact plugs are coplanar with each other, wherein the memory vertical structure and the dummy vertical structure are contacting the pattern structure, and wherein at least one of the dummy vertical structures extend further into the pattern structure than the memory vertical structure in a downward direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. application Ser. No. 17/204,248, filed Mar. 17, 2021, which claims the benefit of Korean Patent Application No. 10-2020-0100116 filed on Aug. 10, 2020, in the Korean Intellectual Property Office, the subject matter of which is hereby incorporated by reference.

BACKGROUND

(1) Embodiments of the present disclosure relate to semiconductor devices and electronic systems

including same.

(2) Contemporary and emerging electronic devices continue a demand for semiconductor devices capable of storing high-capacity data. Accordingly, various technologies and approaches directed to increasing the data storage capacity of semiconductor devices have been investigated. In one approach, semiconductor devices arrange constituent memory cells three-dimensionally instead of two-dimensionally.

SUMMARY

(3) Embodiments of the present disclosure provide semiconductor devices having greater data storage capacity, denser integration density and greater reliability. Other embodiments of the present disclosure provide electronic systems including such semiconductor device(s).

(4) According to an embodiment of the present disclosure, a semiconductor device includes; a lower structure including a peripheral circuit, a lower insulating structure covering the peripheral circuit, and a pattern structure on the lower insulating structure, a stack structure including interlayer insulating layers and horizontal layers alternately stacked on the lower structure, wherein the horizontal layers include gate horizontal layers disposed in a gate region of the stack structure and first insulating horizontal layers disposed in a first insulating region of the stack structure, a memory vertical structure including a portion penetrating the gate horizontal layers in a vertical direction, dummy vertical structures including a portion penetrating the gate horizontal layers in the vertical direction and spaced apart from the memory vertical structure, a first peripheral contact plug including a portion penetrating the first insulating region in the vertical direction and gate contact plugs disposed on gate pads of the gate horizontal layers. The gate contact plugs and the first peripheral contact plugs have upper surfaces disposed at the same level, the memory vertical structure includes a first material different from a material of the dummy vertical structures, the memory vertical structure and the dummy vertical structure contact the pattern structure, and at least one of the dummy vertical structures extends further downwardly into the pattern structure than the memory vertical structure from the upper surface of the pattern structure.

(5) According to an embodiment of the present disclosure, a semiconductor device includes; a lower structure including a peripheral circuit and a pad pattern electrically connected to the peripheral circuit, a stack structure including interlayer insulating layers and horizontal layers alternately stacked on the lower structure, where the stack structure extends from a memory cell array region to a staircase region adjacent to the memory cell array region, the stack structure includes a gate region and an insulating region, and a side surface of the insulating region is at least partially surrounded by the gate region in the staircase region, and the horizontal layers include gate horizontal layers disposed in the gate region and insulating horizontal layers disposed in the insulating region, a memory vertical structure including a portion penetrating the gate horizontal layers in a vertical direction in the memory cell array region, dummy vertical structures including a portion penetrating the gate horizontal layers in the vertical direction in the staircase region, a peripheral contact plug contacting the pad pattern, extending in the vertical direction, and penetrating the insulating region, and gate contact plugs disposed on the gate pads of the gate horizontal layers in the staircase region. The gate contact plugs and the peripheral contact plugs have upper surfaces disposed at the same level, the stack structure includes a first staircase region and a second staircase region arranged in a first direction away from the memory cell array region, and a staircase connection region disposed between the first staircase region and the second staircase region in the staircase region, each of the first and second staircase regions includes a staircase shape extending downwardly by a first height difference in the first direction, an upper surface of the staircase connection region has a shape that is at least one of flattened in the first direction, and having a height difference less than the first height difference in the first direction, a portion of the staircase connection region is an insulating region, and the memory vertical structure includes a material different from a material of the dummy vertical structures.

(6) According to an embodiment of the present disclosure, an electronic system includes; a main

substrate, a semiconductor device on the main substrate, and a controller electrically connected to the semiconductor device on the main substrate. The semiconductor device includes; a lower structure including a peripheral circuit, a lower insulating structure covering the peripheral circuit, and a pattern structure on the lower insulating structure, a stack structure including interlayer insulating layers and horizontal layers alternately stacked on the lower structure, wherein the horizontal layers include gate horizontal layers disposed in the gate region of the stack structure and insulating horizontal layers disposed in the insulating region of the stack structure, a memory vertical structure including a portion penetrating the gate horizontal layers in a vertical direction, dummy vertical structures including a portion penetrating the gate horizontal layers in the vertical direction and spaced apart from the memory vertical structure, a peripheral contact plug including a portion penetrating the insulating region in the vertical direction, and gate contact plugs disposed on gate pads of the gate horizontal layers. The gate contact plugs and the peripheral contact plugs have upper surfaces disposed at the same level, each of the dummy vertical structures has a width greater than a width of each of the memory vertical structures at the same level, the memory vertical structure includes a material different from a material of the dummy vertical structures, the memory vertical structure and the dummy vertical structure extend downwardly from an upper surface of the pattern structure into the pattern structure to contact the pattern structure, and at least one of the dummy vertical structures extends downwardly further into the pattern structure than the memory vertical structure from the upper surface of the pattern structure.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and other aspects, features, and advantages of the present disclosure may be more clearly understood upon consideration of the following detailed description together with the accompanying drawings, in which:
- (2) FIG. 1 is a plan (or top down view) diagram illustrating a semiconductor device according to embodiments of the present disclosure;
- (3) FIGS. 2A, 2B, 2C, 3A, 3B, 3C, 4A, 4B, 4C, 5, 6A, 6B, 7A, 7B, 8, 9A, 9B, 10A, 10B, 10C, 11A, 11B, 11C, 12A, 12B, 13, 14A, 14B, 14C are respective cross-sectional diagrams variously illustrating portion(s) of semiconductor devices according to embodiments of the present disclosure;
- (4) FIGS. 15A, 15B, 16A, 16B, 16C, 17A and 17B are respective and related cross-sectional diagrams illustrating in one example a method of manufacturing a semiconductor device according to embodiments of the present disclosure;
- (5) FIG. 18 is a conceptual diagram illustrating an electronic system including a semiconductor device according to embodiments of the present disclosure;
- (6) FIG. 19 is a perspective diagram illustrating an electronic system including a semiconductor device according to embodiments of the present disclosure; and
- (7) FIG. 20 is a cross-sectional diagram illustrating an electronic system including a semiconductor device according to embodiments of the present disclosure.

DETAILED DESCRIPTION

- (8) Hereinafter, embodiments of the present disclosure will be described with reference to the accompanying drawings. Throughout the written description and drawings, like reference numbers and labels are used to denote like or similar elements and/or features. Throughout the written description certain geometric terms may be used to highlight relative relationships between elements, components and/or features with respect to certain embodiments of the inventive concept. Those skilled in the art will recognize that such geometric terms are relative in nature, arbitrary in descriptive relationship(s) and/or directed to aspect(s) of the illustrated embodiments.

Geometric terms may include, for example: height/width; vertical/horizontal; top/bottom; higher/lower; closer/farther; thicker/thinner; proximate/distant; above/below; under/over; upper/lower; center/side; surrounding; overlay/underlay; etc.

(9) A semiconductor device according to embodiments of the present disclosure will be described in relation to FIGS. 1, 2A, 2B, and 2C, wherein FIG. 1 is a plan view illustrating a semiconductor device 1, FIG. 2A is a cross-sectional diagram taken along line I-I' of FIG. 1, FIG. 2B is a cross-sectional diagram taken along line II-II' of FIG. 1, and FIG. 2C is a cross-sectional diagram taken along line III-III' of FIG. 1.

(10) Referring collectively to FIGS. 1, 2A, 2B, and 2C, the semiconductor device 1 may include a lower structure 3, a stack structure ST, a capping insulating structure 55, memory vertical structures 58, dummy vertical structures 72, separation structures 82, peripheral contact plugs 86p and gate contact plugs 86g.

(11) The semiconductor device 1 may further include a pattern structure 20, as well as a bit line 93, gate connection wirings 94 and a source connection wiring 95.

(12) The semiconductor device 1 may further include a bit line connection pattern 90, a gate connection pattern 91, peripheral contact connection patterns 91a, 91b, and 91c, a source contact connection pattern 91s, and an outer peripheral contact connection pattern 91d.

(13) In some embodiments, the lower structure 3 may include a semiconductor substrate 5, a peripheral circuit PC on the semiconductor substrate 5, and a lower insulating structure 18 covering the peripheral circuit PC on the semiconductor substrate 5. Here, the peripheral circuit PC may include a peripheral device 12 disposed on the semiconductor substrate 5 and a peripheral wiring structure 14 electrically connected to the peripheral device 12 on the semiconductor substrate 5.

(14) The peripheral device 12 may include a transistor including a peripheral source/drain 10 and a peripheral gate 9g. For example, the peripheral gate 9g may be disposed on the active region 7a defined by a device isolation layer 7s on the semiconductor substrate 5, and the peripheral source/drain 10 may be disposed in the active region 7a on both sides of the peripheral gate 9g. The peripheral gate 9g may include a peripheral gate dielectric layer 9a and a peripheral gate electrode 9b stacked in order. The peripheral wiring structure 14 may include conductor layers extending in a vertical direction (e.g., a third or Z direction) as well as horizontal directions (e.g., a first or X direction and a second or Y direction).

(15) In some embodiments, the lower structure 3 may further include peripheral pad patterns 15p electrically connected to the peripheral wiring structure 14 on the peripheral wiring structure 14. Each of the peripheral pad patterns 15p may have a thickness (e.g., measured in the vertical direction relative to the horizontal orientation of the semiconductor substrate 5) greater than a thickness of the gate electrode 9b. The peripheral pad patterns 15p may include a metal material such as tungsten. The peripheral pad patterns 15p may include a first peripheral pad pattern 15p_1, a second peripheral pad pattern 15p_2, a third peripheral pad pattern 15p_3, and an outer peripheral pad pattern 15p_4.

(16) The lower insulating structure 18 may include a first lower insulating layer 18a covering the peripheral wiring structure 14 on the semiconductor substrate 5 and at least partially surrounding side surfaces of the peripheral pad patterns 15p, an etch stop layer 18b on the first lower insulating layer 18a and the peripheral pad patterns 15p, and a second lower insulating layer 18c on the etch stop layer 18b.

(17) The etch stop layer 18b may be formed of a material different from a material of the first and third lower insulating layers 18a and 18c adjacent to the etch stop layer 18b. For example, the etch stop layer 18b may be formed of silicon nitride or a high-k dielectric material, and the first and third lower insulating layers 18a and 18c adjacent to the etch stop layer 18b may be formed of silicon oxide or a low-k dielectric material.

(18) In some embodiments, the lower structure 3 may further include a pattern structure 20.

(19) For example, the pattern structure 20 may include a lower pattern layer 22, a first intermediate

pattern layer **25a** and a second intermediate pattern layer **25b** spaced apart from each other on the lower pattern layer **22**, an upper pattern layer **27** covering the first and second intermediate pattern layers **25a** and **25b** on the lower pattern layer **16**.

(20) In this regard, the lower pattern layer **22** may include a first polysilicon, the first intermediate pattern layer **25a** may include a second polysilicon, and the upper pattern layer **27** may include a third polysilicon. For example, the lower pattern layer **22**, the first intermediate pattern layer **25a**, and the upper pattern layer **27** may include polysilicon having an N-type conductivity.

(21) In an example, the upper pattern layer **27** may contact the lower pattern layer **22** between the first and second intermediate pattern layers **25a** and **25b**, may partially penetrate the second intermediate pattern layer **25b**, and may contact the lower pattern layer **22**. Portions in which the upper pattern layer **27** and the lower pattern layer **22** contacting each other may be referred to as support portions **27s**.

(22) The second intermediate pattern layer **25b** may include an insulating material.

(23) In some embodiments, the lower structure **3** may further include a ground structure. Here, the ground structure may include a ground impurity region **11** formed in the active region **7a** of the semiconductor substrate **5**, a ground wiring structure **14g** electrically connected to the ground impurity region **11** on the ground impurity region **11**, a ground pad pattern **15g** electrically connected to the ground wiring structure **14g** on the ground wiring structure **14g**, and a ground pattern **22v** extending from a lower portion of the lower pattern layer **22** and electrically connected to the ground pad pattern **15g**. The ground impurity region **11** may have N-type conductivity. The ground pad pattern **15g** may be formed of the same material as that of the peripheral pad patterns **15p** on the same level as that of the peripheral pad patterns **15p**.

(24) In some embodiments, the lower structure **3** may further include a first gap-fill insulating layer **29a**, a second gap-fill insulating layer **29b**, and a third gap-fill insulating layer **29c**, penetrating the pattern structure **20**.

(25) For example, the lower structure **3** may further include an outer insulating layer **29d** surrounding an outer surface of the pattern structure **20**. The lower structure **3** may further include an insulating layer **29e** filling a concave portion of an upper surface of the pattern structure **20**. For example, the insulating layer may fill a concave portion of an upper surface of the support portion **27s**.

(26) The stack structure **ST** may extend from the memory cell array region **MCA** to a stair-stepped (or staircase) region **SA** adjacent to the memory cell array region **MCA** on the lower structure **3**.

(27) That is, the stack structure **ST** may be formed in a staircase shape within the staircase region **SA**. For example, in the staircase region **SA**, the stack structure **ST** may include an upper staircase region **IS1**, a first staircase connection region **CS1**, an intermediate stair region **IS2**, a second staircase connection region **CS2**, and a lower staircase region **IS3**, arranged in the first direction (e.g., the X direction) away from the memory cell array region **MCA**.

(28) In the stack structure **ST**, each of the upper, intermediate and lower staircase regions **IS1**, **IS2**, and **IS3** may have a staircase shape decreasing in the first direction, and the upper surface of each of the first and second staircase connection regions **CS1** and **CS2** may have a substantially flattened shape in the first direction or may have a shape in which a height difference thereof may be less than a height difference of the staircases of the upper, intermediate and lower staircase regions **IS1**, **IS2**, and **IS3**.

(29) The stack structure **ST** may include a gate region **ST_G** and an insulating region **ST_I**. In the stack structure **ST**, the insulating region **ST_I** may include a first insulating region **ST_Ia** adjacent to the memory cell array region **MCA**, a second insulating region **ST_Ib** disposed in the first staircase connection region **CS1**, and a second insulating region **ST_Ic** disposed in the second staircase connection region **CS2**, in the second direction (e.g., a Y direction substantially perpendicular to the first direction). In the stack structure **ST**, the gate region **ST_G** may be a region other than the insulating region **ST_I**. Accordingly, the insulating region **ST_1** may be at

least partially surrounded by the gate region ST_G.

(30) In some embodiments, the stack structure ST may include interlayer insulating layers **33** and **43** and horizontal layers **48** alternately stacked on the lower structure **3**. The horizontal layers **48** may include gate horizontal layers **36g** and **46g** and insulating horizontal layers **36i** and **46i**. The gate horizontal layers **36g** and **46g** may include a conductive material, and the insulating horizontal layers **36i** and **46i** may be formed of an insulating material such as silicon oxide.

(31) The gate horizontal layers **36g** and **46g** may be disposed within the gate region ST_G, and the insulating horizontal layers **36i** and **46i** may be disposed within the insulating region ST_I.

(32) For example, the stack structure ST may include a lower stack structure ST1a and an upper stack structure ST2a on the lower stack structure ST1a.

(33) The lower stack structure ST1a may include lower interlayer insulating layers **33** and lower horizontal layers **36** alternately stacked. An uppermost layer of the lower interlayer insulating layers **33** and the lower horizontal layers **36** may be an uppermost lower interlayer insulating layer **33U**, and a lowermost layer may be a lowermost lower interlayer insulating layer **33L**. The uppermost lower interlayer insulating layer **33U** of the lower interlayer insulating layers **33** may have a thickness greater than a thickness of each of the other lower interlayer insulating layers. The lower horizontal layers **36** may include lower gate horizontal layers **36g** and lower insulating horizontal layers **36i**.

(34) The lower insulating horizontal layers **36i** may include first lower insulating horizontal layers **36i_1** in the first insulating region ST_Ia, second lower insulating horizontal layers **36i_2** in the second insulating region ST_Ib, and third lower insulating horizontal layers **36i_3** in the third insulating region ST_Ic.

(35) The upper stack structure ST2a may include upper interlayer insulating layers **43** and lower horizontal layers **46** alternately stacked. An uppermost layer of the upper interlayer insulating layers **43** and the lower horizontal layers **36** may be an uppermost upper interlayer insulating layer **43U**, and a lowermost layer may be a lowermost upper interlayer insulating layer **43L**. The uppermost upper interlayer insulating layer **43U** of the upper interlayer insulating layers **43** may have a thickness greater than a thickness of each of the other upper interlayer insulating layers. The upper horizontal layers **46** may include upper gate horizontal layers **46g** and upper insulating horizontal layers **46i**.

(36) The upper insulating horizontal layers **46i** may include first upper insulating horizontal layers **46i_1** in the first insulating region ST_Ia, and second upper insulating horizontal layers **46i_2** in the second insulating region ST_Ib.

(37) The interlayer insulating layers **33** and **43** may include the lower interlayer insulating layers **33** and the upper interlayer insulating layers **43**. The horizontal layers **48** may include the lower horizontal layers **36** and the upper horizontal layers **46**.

(38) The semiconductor device **1** may further include a capping insulating structure **55** covering a portion of the stack structure ST on the lower structure **3**. The capping insulating structure **55** may have an upper surface coplanar with an upper surface of the stack structure ST and may cover a staircase-shaped portion of the stack structure ST. The capping insulating structure **55** may include a lower capping insulating layer **39** covering a staircase-shaped portion, and an upper capping insulating layer **53** covering a staircase-shaped portion of the upper stack structure ST2a on the lower capping insulating layer **39**.

(39) The memory vertical structure **58** may penetrate the gate region ST_G of the stack structure ST in the memory cell array region MCA. The memory vertical structure **58** may contact the pattern structure **20**.

(40) The semiconductor device **1** may further include vertically stacked first, second, third and fourth upper insulating layers **69**, **76**, **84**, and **88**.

(41) The dummy vertical structures **72** may penetrate the gate region ST_G of the stack structure ST and may extend in the vertical direction in the staircase region SA, and may penetrate the upper

insulating layer **69**. At least one of the dummy vertical structures **72** may penetrate the capping insulating structure **55** disposed between the first upper insulating layer **69** and the stack structure **ST**.

(42) For example, the dummy vertical structures **72** may include silicon oxide or a low-k dielectric layer.

(43) The separation structures **82** may penetrate the stack structure **ST** and may extend in the vertical direction. The separation structures **82** may extend in the vertical direction from a portion penetrating the stack structure **ST** and may penetrate the second upper insulating layer **76**.

(44) The separation structures **82** may include main separation structures **82a** isolating and spacing the stack structures **ST** in the second direction, and auxiliary separation structures **82b** each having a length less than a length of each of the main separation structures **82a** between the main separation structures **82a**. The separation structures **82** may include silicon oxide.

(45) The semiconductor device **1** may further include a dam structure **80** surrounding each of the second and third insulating regions **ST_Ib** and **ST_Ic**. In an example, the dam structure **80** may include a material different from a material of the separation structures **82**. For example, the dam structure **80** may include a material different from the material of the separation structures **82**, at least one of polysilicon and silicon nitride, for example.

(46) The peripheral contact plugs **86p** may have upper surfaces disposed on substantially the same level. Each of the peripheral contact plugs **86p** may include a portion penetrating the first, second and third upper insulating layers **69**, **76**, and **84** and extending downwardly to penetrate the insulating region **ST_I**. For example, the peripheral contact plugs **86p** may include a first peripheral contact plug **86p_1** including a portion penetrating the first insulating region **ST_Ia** in the vertical direction, a second peripheral contact plug **86p_b** including a portion penetrating the second insulating region **ST_Ib** in the vertical direction, and a third peripheral contact plug **86p_3** including a portion penetrating the third insulating region **ST_Ic** in the vertical direction.

(47) The first peripheral contact plug **86p_1** may penetrate the first gap-fill insulating layer **29a** disposed in a lower portion of the first insulating region **ST_Ia** and may extend downwardly to contact the first pad pattern **15p_1**. The second peripheral contact plug **86p_2** may penetrate the second gap-fill insulating layer **29b** disposed in a lower portion of the second insulating region **ST_Ib** and may extend downwardly to contact the second pad pattern **15p_2**. The third peripheral contact plug **86p_3** may penetrate the third gap-fill insulating layer **29c** disposed in a lower portion of the third insulating region **ST_Ic** and may extend downwardly to contact the third pad pattern **15p_3**.

(48) The semiconductor device **1** may further include a source contact plug **86s** penetrating the first, second and third upper insulating layers **69**, **76**, and **84** and the capping insulating structure **55** from an external side of the pattern structure **20**, extending downwardly, penetrating the outer peripheral contact plug **86op** to contact the outer pad pattern **15p_4**, the first, second and third upper insulating layers **69**, **76**, and **84**, and the capping insulating structure **55** extending into the pattern structure **20** to contact the lower pattern layer **22** of the pattern structure **20**.

(49) The peripheral contact plugs **86p**, the source contact plug **86s**, and the outer peripheral contact plug **86op** may have upper surfaces disposed on substantially the same level.

(50) The gate contact plugs **86g** may contact and be electrically connected to the lower gate pads **36p** of the lower gate horizontal layers **36g** and the upper gate pads **46p** of the upper gate horizontal layers **46g**. The gate contact plugs **86g**, the peripheral contact plugs **86p**, the source contact plug **86s**, and the outer peripheral contact plug **86op** may include the same conductive material. The gate contact plugs **86g**, the peripheral contact plugs **86p**, the source contact plug **86s**, and the outer peripheral contact plug **86op** may have upper surfaces disposed on substantially the same level.

(51) The semiconductor device **1** may include contact spacer layers **74s_1**, **74s_2**, **74s_3**, **74s_s**, and **74s_o** covering a portion of side surfaces of each of the peripheral contact plugs **86p**, the source contact plug **86s**, and the outer peripheral contact plug **86op**. For example, the contact

spacer layers **74s_1**, **74s_2**, **74s_3**, **74s_s**, and **74s_o** may include a first peripheral contact spacer layer **74s_1** covering a portion of a side surface of the first peripheral contact plug **86p_1**, the second peripheral contact spacer layer **74s_2** covering a portion of the side surface of the second peripheral contact plug **86p_2**, the third peripheral contact spacer layer **74s_3** covering a portion of the side surface of the third peripheral contact plug **86p_3**, the source contact spacer layer **74s_s** covering a portion of the side surface of the source contact plug **86s**, and the outer peripheral contact spacer layer **74s_o** covering a portion of the side surface of the outer peripheral contact plug **86op**. Upper ends of the contact spacer layers **74s_1**, **74s_2**, **74s_3**, **74s_s**, and **74s_o** may be disposed at the same level. The contact spacer layers **74s_1**, **74s_2**, **74s_3**, **74s_s**, and **74s_o** may be referred to as insulating contact spacer layers.

(52) The semiconductor device **1** may further include bit lines **93**, gate connection wirings **94**, and source connection wirings **95**. In an example, the bit lines **93**, the gate connection wirings **94**, and the source connection wiring **95** may be disposed on the fourth upper insulating layer **88**.

(53) The semiconductor device **1** may include bit line connection patterns **90**, gate connection patterns **91g**, first peripheral contact connection patterns **91a**, second peripheral contact connection patterns **91b**, a third peripheral contact connection pattern **91c**, a source contact connection pattern **91s**, and an outer contact connection pattern **91d**.

(54) The bit line connection patterns **90** may electrically connect the bit lines **93** to the memory vertical structure **58** between the bit lines **93** and the memory vertical structure **58**. The gate connection patterns **91g** may electrically connect the gate connection wirings **94** to the gate contact plugs **86g** between the gate connection patterns **94** and the gate contact plugs **86g**.

(55) The first peripheral contact connection pattern **91a** may electrically connect the first peripheral contact plug **86p_1** to the bit line **93** between the first peripheral contact plug **86p_1** and the bit line **93**. The second and third peripheral contact connection patterns **91b** and **91c** may electrically connect the second and third peripheral contact plugs **86p_2** to **86p_3** to the gate connection wirings **94** between the second and third peripheral contact plugs **86p_2** to **86p_3** and the gate connection wirings **94**.

(56) The source contact connection pattern **91s** may electrically connect the source contact plug **86s** to the source connection wiring **95**, and the outer contact connection pattern **91d** may electrically connect the outer peripheral contact plug **86op** to the source connection wiring **95**.

(57) In the written description that follows, various examples and modifications (e.g., modified examples) of the foregoing will be described with continued reference to the embodiments illustrated in FIGS. **1**, **2A**, **2B** and **2C**.

(58) Further, an example of the gate horizontal layers **36g** and **46g** and the memory vertical structure **58** will be described with reference to FIG. **3A**, wherein FIG. **3A** is an enlarged cross-sectional diagram illustrating the portion “**A1**” identified in FIG. **2C**.

(59) Referring to FIG. **3A**, each of the gate horizontal layers **36g** and **46g** may include a first gate layer **47a** and a second gate layer **47b**. The first gate layer **47a** may cover upper and lower surfaces of the second gate layer **47b** and may be interposed between the second gate layer **47b** and the memory vertical structure **58**.

(60) For example, the first gate layer **47a** may include a dielectric material, and the second gate layer **47b** may include a conductive material. For example, the first gate layer **47a** may include a high-k dielectric such as **AlO**, and the second gate layer **47b** may include a conductive material such as **TiN**, **WN**, **Ti**, or **W**.

(61) In other examples, the first gate layer **47a** may include a first conductive material (e.g., **TiN** or **W**), and the second gate layer **47b** may include a second conductive material different from the first conductive material (e.g., **Ti** or **W**).

(62) In still other examples, each of the first and second gate layers **47a** and **47b** may be formed of a doped polysilicon, a metal-semiconductor compound (e.g., **TiSi**, **TaSi**, **CoSi**, **NiSi** or **WSi**), a metal nitride (e.g., **TiN**, **TaN** or **WN**) or a metal (e.g., **Ti** or **W**).

(63) In some embodiments, the lowermost first lower gate horizontal layer **33g_La** of the gate horizontal layers **36g** and **46g** may be lower erase control gate electrodes, and the first lower gate horizontal layer **33g_La** on the second lower gate horizontal layer **33g_Lb** may be a ground select gate electrode. A first upper gate horizontal layer **46g_Ua** of the gate horizontal layers **36g** and **46g** may be a string select gate electrode, and a second upper gate horizontal layer **46g_Ub** on the first upper gate horizontal layer **46g_Ua** may be an upper erase control gate electrode. Each of the first lower gate horizontal layer **33g_La** and the second upper gate horizontal layer **46g_Ub** may include one or more vertically stacked layers (**Z**).

(64) In an example, at least a portion of intermediate gate horizontal layers **36M** and **46M** of the gate horizontal layers **36g** and **46g**, disposed between the second lower gate horizontal layer **36g_Lb** and the first upper gate horizontal layer **46g_Ua**, may be word lines.

(65) The memory vertical structure **58** may include an insulating core pattern **64**, a channel layer **62** covering a side surface and a bottom surface of the insulating core pattern **64**, a data storage structure **60** disposed on an outer surface and a bottom surface of the channel layer **62**, and a pad pattern **66** contacting the channel layer **62** on the insulating core pattern **64**.

(66) The insulating core pattern **64** may include silicon oxide. The channel layer **62** may include polysilicon. The pad pattern **66** may include at least one of a doped polysilicon, a metal nitride (e.g., TiN, or the like), a metal (e.g., W or the like), and a metal-semiconductor compound (e.g., TiSi or the like).

(67) The data storage structure **60** may include a first dielectric layer **60a**, a second dielectric layer **60c**, and a data storage layer **60b** disposed between the first dielectric layer **60a** and the second dielectric layer **60c**. The second dielectric layer **60c** may contact the channel layer **62**, and the data storage layer **60b** may be spaced apart from the channel layer **62**. The first dielectric layer **60a** may include silicon oxide or silicon oxide doped with impurities. The second dielectric layer **60c** may include at least one of silicon oxide and a high-k material. The data storage layer **60b** may include regions which may store data in the semiconductor device such as a flash memory device. For example, the data storage layer **60b** may include a material for trapping a charge, such as silicon nitride, for example.

(68) The first intermediate pattern layer **25a** may penetrate the data storage structure **60** to contact the channel layer **62**. Accordingly, the data storage structure **60** may be divided into a lower portion **60L** and an upper portion **60U** by the first intermediate pattern layer **25a**.

(69) A side surface of the memory vertical structure **58** may have an inflection portion **58V** in a region in which the upper stack structure **ST1a** and the lower stack structure **ST1a** are adjacent to each other. For example, the memory vertical structure **58** may include a lower vertical portion **58L** disposed in the lower stack structure **ST1a** and an upper vertical portion **58U** disposed in the upper stack structure **ST2a**. A width of the lower region of the upper vertical portion **58U** may be less than a width of the upper region of the lower vertical portion **58L**. Due to this difference in widths, a side surface of the memory vertical structure **58** may have a bent portion, the inflection portion **58V**, in a region in which the upper vertical portion **58U** and the lower vertical portion **58L** are adjacent to each other.

(70) In an example, the memory vertical structure **58** may extend from the upper surface **20s** of the pattern structure **20** into the pattern structure **20** by a first depth **D1a**. Accordingly, the memory vertical structure **58** may penetrate the upper pattern layer **27** and the first intermediate pattern layer **25a** in order and may extend into the lower pattern layer **22**.

(71) An example of the dummy vertical structures **72** will be described with reference to FIGS. 3B and 3C, wherein FIG. 3B is an enlarged cross-sectional diagram illustrating the portion “**A2a**” identified in FIG. 2C, and FIG. 3C is an enlarged cross-sectional diagram illustrating the portion “**A2b**” identified in FIG. 2C.

(72) Referring to FIGS. 3B and 3C, upper surfaces of the dummy vertical structures **72** may be disposed on a level higher than a level of upper surfaces of the memory vertical structures **58**. (See,

e.g., FIG. 3A). The dummy vertical structures **72** may be formed of silicon oxide. The memory vertical structure **58** may include a material different from a material of the dummy vertical structures **72**, a material of the channel layer **62** and a material of the pad pattern **66**, for example. (73) Each of the dummy vertical structures **72** may have a width greater than that of the memory vertical structure **58** on one of levels.

(74) The dummy vertical structures **72** may penetrate the upper pattern layer **27** and the second intermediate pattern layer **25b** of the pattern structure **20** and may extend into the lower pattern layer **22**. In an example, the second intermediate pattern layer **25b** may include a first layer **26a**, a second layer **26b**, and a third layer **26c** stacked in order. The first and third layers **25a** and **25c** of the second intermediate pattern layer **25b** may be a silicon oxide layer, and the second layer **26b** may be a silicon nitride or polysilicon layer.

(75) A depth by which the dummy vertical structures **72** extend into the lower pattern layer **22** may be greater than a first depth **D1a** by which the memory vertical structure **58** extends from the upper surface **20s** of the pattern structure **20** into the pattern structure **20**. (See, e.g., FIG. 3A). For example, the dummy vertical structures **72** may include a first dummy vertical structure **72a** (reference FIG. 3B) adjacent to the memory cell array region MCA and a second dummy vertical structure **72b** (reference FIG. 3C) spaced apart from the memory cell array region MCA.

(76) The first dummy vertical structure **72a** may extend from the upper surface **20s** of the pattern structure **20** into the pattern structure **20** by a second depth **D1b** greater than the first depth **D1a**. (See, e.g., FIG. 3A). The second dummy vertical structure **72b** may extend from the upper surface **20s** of the pattern structure **20** into the pattern structure **20** by a third depth **D1c** greater than the first depth **D1a**. Here, the third depth **D1c** may be greater than the second depth **D1b**.

(77) Embodiments of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3** and the first, second and third peripheral contact spacer layers **74s_1** and **74s_2**, and **74s_3** will be described with reference to FIGS. 4A, 4B, and 4C. Here, FIG. 4A is an enlarged cross-sectional diagram illustrating the portion “**B1a**” identified in FIG. 2A, FIG. 4B is an enlarged cross-sectional diagram illustrating the portion “**B1b**” identified in FIG. 2B, and FIG. 4C is an enlarged cross-sectional diagram illustrating the portion “**B1c**” identified in FIG. 2C.

(78) Referring to FIGS. 4A, 4B, and 4C, each of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3** may include a plug conductive pattern **86c**, a first conductive liner layer **86b** covering a side surface and a bottom surface of the plug conductive pattern **86c**, and a second conductive liner layer **86a** covering an external surface and a bottom surface of the first conductive liner layer **86b**. For example, the second conductive liner layer **86a** may include a metal such as Ti, and the first conductive liner layer **86b** may include a metal nitride such as TiN, and the plug conductive pattern **86c** may include a metal such as W.

(79) A width of each of the first, second and third peripheral contact spacer layers **74s_1**, **74s_2**, and **74s_3** in a horizontal direction may be less than a width of each of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3**.

(80) Referring to FIG. 4A, the first peripheral contact spacer layer **74s_1** may extend from the upper surface **15s** of the first peripheral pad pattern **15p_1** into the first peripheral pad pattern **15p_1** by a first depth **D2a**.

(81) Referring to FIG. 4B, the second peripheral contact spacer layer **74s_2** may extend from the upper surface **15s** of the second peripheral pad pattern (**15p_2**) into the second peripheral pad pattern **15p_2** by a second depth **D2b** greater than the first depth **D2a**. Referring to FIG. 4C, the third peripheral contact spacer layer **74s_3** may extend from the upper surface **15s** of the third peripheral pad pattern **15p_3** into the third peripheral pad pattern **15p_3** by a third depth **D2c** greater than the second depth **D2b**.

(82) In some embodiments, each of the first, second and third depths **D2a**, **D2b**, and **D2c** may be greater than the thickness **t1** of the etch stop layer **18b**.

(83) The first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3** may extend

further into the first, second and third peripheral pad patterns **15p_1**, **15p_2**, and **15p_3** than the first, second and third peripheral contact spacer layers **74s_1**, **74s_2**, and **74s_3**, respectively. For example, the first peripheral contact plug **86p_1** may extend further than a lower surface of the first peripheral contact spacer layer **74s_1** into the first peripheral pad pattern **15p_1** by a first depth **D3a**. The second peripheral contact plug **86p_2** may extend further into the second peripheral pad pattern **15p_2** than a lower surface of the second peripheral contact spacer layer **74s_2** by a second depth **D3b**, and the third peripheral contact plug **86p_3** may extend further into the third peripheral pad pattern **15p_3** than the lower surface of the third peripheral contact spacer layer **74s_3** by a third depth **D3c**.

(84) In some embodiments, the first, second and third depths **D3a**, **D3b**, and **D3c** of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3** may be less than the first, second and third depths **D2a**, **D2b**, and **D2c** of the first, second and third peripheral contact spacers **74s_1**, **74s_2**, and **74s_3**.

(85) For example, the first, second and third depths **D3a**, **D3b**, and **D3c** of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3** may be less than the respective thicknesses of the horizontal layers **48**.

(86) Hereafter, embodiments of the source contact plug **86s** and the source contact spacer layer **74s_s** will be described with reference to FIG. 5, wherein FIG. 5 is an enlarged cross-sectional diagram illustrating the portion “C” identified in FIG. 2B.

(87) Referring to FIG. 5, the source contact plug **86s** may include the plug conductive pattern **86c**, and the first conductive liner layer **86b**, and the second conductive liner layer **86a**, that are constitutively and respectively the same as the first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3**.

(88) For example, a portion of the second conductive liner layer **86a** contacting the lower pattern layer **22** may be formed of a metal-semiconductor compound layer **86d**. The metal-semiconductor compound layer **86d** may include metal silicide such as TiSi.

(89) The source contact spacer layer **74s_s** may extend from the upper surface **20s** of the pattern structure **20** into the pattern structure **20** by a first depth **D4a**. The source contact spacer layer **74s_s** may penetrate the upper pattern layer **27** and the second intermediate pattern layer **25b** and may extend into the lower pattern layer **22**.

(90) The source contact plug **86s** may extend further than the source contact spacer layer **74s_s** into the lower pattern layer **22** by a first depth **D4b**. Referring to FIGS. 4A, 4B and 4C, the first depth **D4b** of the source contact plug **86s** may be greater than the first, second and third depths **D3a**, **D3b**, and **D3c** of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3**. The first depth **D4b** of the source contact plug **86s** may be greater than the thickness **t1** of the etch stop layer **18b**. (Sec. e.g., FIG. 4A).

(91) Modified examples of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, and **86p_3** and the first, second and third peripheral contact spacer layers **74s_1**, **74s_2**, and **74s_3** previously described with reference to FIGS. 4A, 4B and 4C will be described with reference to FIGS. 6A and 6B, wherein FIG. 6A is an enlarged cross-sectional diagram illustrating the portion “B1a” identified in FIG. 4A, the portion “B1b” identified in FIG. 4B and the portion B1c identified in FIG. 4C. Here, FIG. 6B is an enlarged cross-sectional diagram illustrating portions “D1a,” “D2a,” and “D3a” of the portion “B1a” identified in FIG. 6A.

(92) Referring to FIGS. 6A and 6B, each of the first, second and third peripheral contact plugs **86p_1a**, **86p_2a**, and **86p_3a** may have first and second plug side surfaces **86p_s1** and **86p_s2** opposing each other, and in each of the first, second and third peripheral contact plugs **86p_1a**, **86p_2a** and **86p_3a**, the second plug side surface **86p_s2** may be more adjacent to the insulating horizontal layers **36i** and **46i** than the first plug side surface **86p_s1**. Each of the first, second and third peripheral contact spacer layers **74s_1a**, **74s_2a**, and **74s_3a** may include a first spacer portion **76s_p1** covering the first plug side surface **86p_s1** of each of the first, second and third

peripheral contact plugs **86p_1a**, **86p_2a**, and **86p_3a**, and a second spacer portion **76s_p2** covering the second plug side surface **86p_s2** of each of the first, second and third peripheral contact plugs **86p_1a**, **86p_2a**, and **86p_3a**. Any one of the first insulating horizontal layers **36i** and **46i** may include a first horizontal portion **46i_1a** facing the first plug side surface **86p_s1** and a second horizontal portion **46i_1b** facing the second plug side surface **86p_s2**. The first spacer portion **76s_p1** may dispose between the first plug side surface **86p_s1** and the first horizontal portion **46i_1a**. The second spacer portion **76s_p2** may dispose between the second plug side surface **86p_s2** and the second horizontal portion **46i_1b**. The width of the first spacer portion **76s_p1** in a horizontal direction may be greater than a width of the second spacer portion **76s_p2** in the horizontal direction.

(93) Modified examples of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, **86p_3** and the first, second and third peripheral contacts spacer layers **74s_1**, **74s_2**, and **74s_3** will be described with reference to FIGS. 7A and 7B, wherein FIG. 7A is an enlarged cross-sectional diagram illustrating the portion “**B1a**” identified in FIG. 4A, the portion “**B1b**” identified in FIG. 4B and the portion “**B1c**” identified in FIG. 4C. Here, FIG. 7B is an enlarged cross-sectional diagram illustrating portions “**D1b**,” “**D2b**,” and “**D3b**” of the portion “**B1a**” identified in FIG. 7A.

(94) Referring to FIGS. 7A and 7B, the first, second and third peripheral contact spacer layers **74s_1**, **74s_2**, and **74s_3** previously described with reference to FIGS. 4A, 4B and 4C may not be provided.

(95) The first peripheral contact plug **86p_1b** may include a lower plug portion **86p_1bL** and an upper plug portion **86p_1bU** having a width greater than that of the lower plug portion **86p_1bL** on the lower plug portion **86p_1bL**. The upper plug portion **86p_1bU** may contact the first upper insulating horizontal layer **46i_1**, disposed on an upper portion, of the insulating horizontal layers **36i** and **46i** adjacent to the first peripheral contact plug **86p_1b** and may be disposed on a level higher than a level of the first upper insulating horizontal layer **46i_1** disposed on an upper portion. In the first upper insulating horizontal layer **46i_1**, a minimum thickness of a portion of the first upper insulating horizontal layer **46i_1** contacting the upper plug portion **86p_1bU** may be less than a thickness of the other portion.

(96) In the first peripheral contact plug **86p_1b**, the upper plug portion **86p_1bU** may include a first portion **86pc** overlapping the lower plug portion **86p_1bL** and a second portion **86pa** overlapping the insulating horizontal layers **36i** and **46i**.

(97) The second peripheral contact plug **86p_2b** may include a lower plug portion **86p_2bL** and an upper plug portion **86p_2bU** having a width greater than that of the lower plug portion **86p_2bL** on the lower plug portion **86p_2bL**. The upper plug portion **86p_2bU** may contact the second upper insulating horizontal layer **46i_2** disposed on the upper portion of the insulating horizontal layers **36i** and **46i** adjacent to the second peripheral contact plug **86p_2b** and may be disposed on a level higher than a level of the second upper insulating horizontal layer **46i_2** disposed on an upper portion.

(98) The third peripheral contact plug **86p_3b** may include a lower plug portion **86p_3bL** and an upper plug portion **86p_3bU** having a width greater than that of the lower plug portion **86p_3bL** on the lower plug portion **86p_3bL**. The upper plug portion **86p_3bU** may contact the third lower insulating horizontal layer **36i_3**, disposed in an upper portion, of the insulating horizontal layers **36i** and **46i** adjacent to the third peripheral contact plug **86p_3b** and may be disposed on a level higher than a level at which the third lower insulating horizontal layer **36i_3** is disposed in an upper portion.

(99) In the first peripheral contact plug **86p_1b**, a vertical height of the upper plug portion **86p_1bU** may be less than a height of the lower plug portion **86p_1bL**. In the third peripheral contact plug **86p_3b**, a vertical height of the upper plug portion **86p_3bU** may be greater than a height of the lower plug portion **86p_3bL**. A vertical height of the upper plug portion **86p_1bU** of the first peripheral contact plug **86p_1b** may be less than a height of the upper plug portion

86p_3bU of the third peripheral contact plug **86p_3b**.

(100) Referring to FIGS. 7A and 7B, a central axis of each of the upper plug portions **86p_1bU**, **86p_2bU**, and **86p_3bU** may be vertically aligned with a central axis of each of the lower plug portions **86p_1bL**, **86p_2bL**, and **86p_3bL**.

(101) Hereafter, modified examples of the upper plug portion **86p_1bU**, **86p_2bU**, and **86p_3bU** of FIGS. 7A and 7B will be described with reference to FIG. 8, wherein FIG. 8 is an enlarged cross-sectional diagram corresponding to the enlarged cross-sectional of FIG. 7B. Here, the term “central axis” denotes an arbitrary axis crossing a center between both side surfaces of an element or region.

(102) Referring to FIGS. 7A and 8, each of the upper plug portions **86p_1bU**, **86p_2bU**, and **86p_3bU** illustrated in FIG. 7A may be modified to have a central axis that is misaligned with a central axis of the lower plug portions **86p_1bL**, **86p_2bL**, and **86p_3bL**. For example, in the first peripheral contact plug **86p_1b'**, a first central axis **Cz_1a** of the upper plug portion **86p_1bU'** may be misaligned with a second central axis **Cz_1b** of the lower plug portion **86p_1bL**.

(103) Modified examples of the first, second and third peripheral contact plugs **86p_1**, **86p_2**, **86p_3** and the first, second and third peripheral contact spacer layers **74s_1**, **74s_2**, and **74s_3** described with reference to FIGS. 4A, 4B and 4C will be described with reference to FIGS. 9A and 9B, wherein FIG. 9A is an enlarged diagram illustrating the portion “B1b” identified in FIG. 4B, the portion “B1b” identified in FIG. 4B and the portion “B1c” identified in FIG. 4C. Here, FIG. 9B is an enlarged diagram illustrating portions “D1c,” “D2c,” and “D3c” of the portion “B1a” of FIG. 9A.

(104) Referring to FIGS. 9A and 9B, the first, second and third peripheral contact plugs **86p_1c**, **86p_2c**, and **86p_3c** may include lower plug portions **86p_1cL**, **86p_2cL**, and **86p_3cL** having a first width, and upper plug portions **86p_1cU**, **86p_2cU**, and **86p_3cU** having a second width greater than the first width and disposed on the lower plug portions **86p_1cL**, **86p_2cL**, and **86p_3cL**. The upper plug portions **86p_1cU**, **86p_2cU**, and **86p_3cU** may be disposed on the same level as that of the upper plug portions **86p_1bU**, **86p_2bU**, and **86p_3bU**, as described, for example, in FIG. 7A.

(105) A first central axis **Cz_2a** of each of the upper plug portions **86p_1cU**, **86p_2cU**, and **86p_3cU** may not be aligned with a second central axis **Cz_2b** of each of the lower plug portions **86p_1cL**, **86p_2cL**, and **86p_3cL**.

(106) In the first, second and third peripheral contact plugs **86p_1c**, **86p_2c**, and **86p_3c**, each of the lower plug portions **86p_1cL**, **86p_2cL**, and **86p_3cL** may have a first lower side surface **86p_s1bL** and a second lower side surface **86p_s2bL** opposing each other, and each of the upper plug portions **86p_1cU**, **86p_2cU**, and **86p_3cU** may have a first upper side surface **86p_s1bU** and a second upper side surface **86p_s2bU** opposing each other. The first lower side **86p_s1bL** and the first upper side **86p_s1bU** may be aligned with each other, and the second upper side **86p_s2bU** may not be aligned with the second lower side **86p_s2bL**. Accordingly, the first, second and third peripheral contact plugs **86p_1c**, **86p_2c**, and **86p_3c** may have asymmetric side surfaces.

(107) The first, second and third peripheral contact spacer layers **74s_1b**, **74s_2b**, and **74s_3b** may cover the first lower side **86p_s1bL** and the first upper side **86p_s1bU** of the first, second and third peripheral contact plugs **86p_1c**, **86p_2c**, and **86p_3c**. In the first, second and third peripheral contact plugs **86p_1c**, **86p_2c**, and **86p_3c**, the second lower side surface **86p_s2bL** may contact the insulating horizontal layers **36i** and **46i**. The upper plug portions **86p_1cU**, **86p_2cU**, and **86p_3cU** may include portions overlapping the insulating horizontal layers **36i** and **46i**.

(108) A modified example of the semiconductor device **1** will now be described with reference to FIGS. 10A, 10B and 10C, wherein FIG. 10A is a cross-sectional diagram taken along line I-I', FIG. 10B is a cross-sectional diagram taken along line II-II' and FIG. 10C is an enlarged cross-sectional diagram illustrating the portion “E” identified in FIG. 10A.

(109) Referring to FIGS. 10A, 10B, and 10C, the lower structure **3** of the semiconductor device **1** may further include buffer pad patterns **19p_1**, **19p_2**, **19p_3**, and **19p_4** contacting the peripheral

pad patterns **15p** on the peripheral pad patterns **15p**. The ground pattern **22V** of FIG. 2B may be replaced with a source buffer pad pattern **19s** formed of the same material as that of the buffer pad patterns **19p_1**, **19p_2**, **19p_3**, and **19p_4**. The source buffer pad pattern **19s** may be electrically connected to the source pad pattern **15sp** disposed in a lower portion of the source buffer pad pattern **19s**. A source wiring structure **14s** electrically connected to the source pad pattern **15sp** may be disposed in a lower portion of the source pad pattern **15sp**. A source control element **11s** of the peripheral circuit PC electrically connected to the source wiring structure **14s** may be disposed.

(110) In some embodiments, upper surfaces of the buffer pad patterns **19p_1**, **19p_2**, **19p_3**, and **19p_4** and the source buffer pad pattern **19s** may be disposed at substantially the same level as a level of the upper surface of the lower insulating structure **18**.

(111) For example, when the source buffer pad pattern **19s** is disposed, the source contact plug **86s** of FIG. 2B may not be provided, and the source connection wiring **95** of FIG. 2B may be replaced with an outer element connection wiring **96**.

(112) Thus, the peripheral contact plugs **86p** of FIGS. 2A and 2B and the outer peripheral contact plug **86op** of FIG. 2B may be respectively modified to a peripheral contact plugs **186p** and an outer peripheral contact plugs **186op** contacting the buffer pad patterns **19p_1**, **19p_2**, **19p_3**, and **19p_4**. Here, the contact spacer layers **74s_1**, **74s_2**, **74s_3**, and **74s_o** described with reference to FIGS. 2A and 2B may be modified to contact spacer layers **174s_1**, **174s_2**, **174s_3**, and **174s_o** contacting the buffer pad patterns **19p_1**, **19p_2**, **19p_3**, and **19p_4**. For example, the peripheral contact plugs **186p** may include a first peripheral contact plug **186p_1** penetrating the first insulating region ST_Ia to contact the first buffer pad pattern **19p_1**, a second peripheral contact plug **186p_2** penetrating the second insulating region ST_Ib to contact the second buffer pad pattern **19p_2**, and a third peripheral contact plug **186p_3** penetrating the third insulating region ST_Ic to contact the second buffer pad pattern **19p_3**.

(113) The shape of portions of the first peripheral contact plug **186p_1** and the first contact spacer layer **174s_1** contacting the first buffer pad pattern **19p_1** may be substantially the same as a shape of portions of the first peripheral contact plug **86p_1** and the first contact spacer layer **74s_1** contacting the first peripheral pad pattern **15p_1** described with reference to FIG. 4A. The shape of portions of the second peripheral contact plug **186p_2** and the second contact spacer layer **174s_2** contacting the second buffer pad pattern **19p_2** may be substantially the same as a shape of portions of second peripheral contact plug **86p_2** and the second contact spacer layer **74s_2** in contact second peripheral pad pattern **15p_2** described with reference to FIG. 4B. The shape of portions of the third peripheral contact plug **186p_3** and the second contact spacer layer **174s_3** contacting the third buffer pad pattern **19p_3** may be substantially the same as a shape of portions of third peripheral contact plug **86p_3** and the third contact spacer layer **74s_3** contacting the third peripheral pad pattern **15p_3** described with reference to FIG. 4C.

(114) In some embodiments, the first, second and third peripheral contact plugs **186p_1**, **186p_2**, **186p_3** and the third contact spacer layers **174s_1**, **174s_2**, and **174s_3** contacting the first, second and third peripheral contact plugs **186p_1**, **186p_2**, **186p_3** may be variously modified. For example, the first, second and third peripheral contact plugs **186p_1**, **186p_2**, **186p_3** and the first, second and third contact spacer layers **174s_1**, **174s_2** and **174s_3** which may be modified as illustrated in FIGS. 11A, 11B and 11C.

(115) In one modified example of FIG. 11A, the first, second and third peripheral contact plugs **186p_1**, **186p_2** and **186p_3** and the first, second and third contact spacer layers **174s_1**, **174s_2**, and **174s_3** in FIGS. 10A and 10B may be modified similarly to the first, second and third peripheral contact plugs **86p_1a**, **86p_2a**, and **86p_3a** and the first, second and third contact spacer layers **74s_1a**, **74s_2a**, and **74s_3a** of FIGS. 6A and 6B. Here, FIG. 11A is an enlarged diagram illustrating a portion in which the first peripheral contact plug **186p_1a** and the first contact spacer layer **174s_1a** contacting the first buffer pad pattern **19p_1**.

(116) In another modified example of FIG. 11B, the first, second and third contact spacer layers

174s_1, **174s_2**, and **174s_3** in FIGS. **10A** and **10B** may not be provided, and the first, second and third peripheral contact plugs **186p_1**, **186p_2**, and **186p_3** in FIGS. **10A** and **10B** may be modified in the same manner as the first, second and third peripheral contact plugs **86p_1a**, **86p_2a**, and **86p_3a** of FIGS. **7A**, **7B** and **8**. Here, FIG. **11B** is an enlarged diagram illustrating a portion in which the first peripheral contact plug **186p_1b** contacting the first buffer pad pattern **19p_1**.

(117) In another modified example of FIG. **11C**, the first, second and third peripheral contact plugs **186p_1**, **186p_2** and **186p_3** and the first, second and third contact spacer layers **174s_1**, **174s_2** and **174s_3** illustrated in FIGS. **10A** and **10B** may be modified in the same manner as the first, second and third peripheral contact plugs **86p_1c**, **86p_2c** and **86p_3c** and the first, second and third contact spacer layers **74s_1b**, **74s_2b**, and **74s_3b** of FIGS. **9A** and **9B**. Here, FIG. **11C** is an enlarged diagram illustrating a portion in which the first contact spacer layer **174s_1b** contacting the first buffer pad pattern **19p_1**.

(118) Hereafter, the modified examples of the buffer pad patterns **19p_1**, **19p_2**, **19p_3**, and **19p_4** and the source pad pattern **19s** of FIGS. **10A**, **10B** and **10C** will be described with reference to FIGS. **12A**, **12B**, and **12C**, wherein FIG. **12A** is a cross-sectional diagram taken along line I-I', FIG. **12B** is a cross-sectional diagram taken along line II-II', and FIG. **13** is an enlarged cross-sectional diagram illustrating portion "F" identified in FIG. **12A**.

(119) Referring to FIGS. **12A**, **12B**, and **13**, the buffer pad patterns **19p_1**, **19p_2**, **19p_3**, and **19p_4** and the source buffer pad pattern **19s** described in FIGS. **10A**, **10B**, and **10C** may be modified to the buffer pad patterns **131p_1**, **131p_2**, **131p_3**, and **131p_4** and the source buffer pad pattern **131s** illustrated in FIGS. **12A**, **12B**, and **12C**, respectively. For example, the buffer pad patterns **131p_1**, **131p_2**, **131p_3**, and **131p_4** may contact the peripheral pad patterns **15p**, may extend upwardly, and may penetrate the first, second and third gap-fill insulating layers **29a**, **29b**, and **29c**, and the source buffer pad pattern **131s** may contact the source pad pattern **15s**, may extend upwardly, and may penetrate the pattern structure **20**. Upper surfaces of the buffer pad patterns **131p_1**, **131p_2**, **131p_3**, and **131p_4** and the source buffer pad patterns **131s** may be disposed on the same level as a level of the upper surface of the pattern structure **20**.

(120) The peripheral contact plugs **186p** and the outer peripheral contact plug **186op** described in FIGS. **10A** and **10B** may be modified to the peripheral contact plugs **286p** and the outer peripheral contact plug **286op** contacting the buffer pad patterns **131p_1**, **131p_2**, **131p_3**, and **131p_4**, as described with reference to FIGS. **10A** and **10B**. The contact spacer layers **174s_1**, **174s_2**, **174s_3**, and **174s_o** described with reference to FIGS. **10A** and **10B** may be modified to the contact spacer layers **274s_1**, **274s_2**, **274s_3** and **274s_o** contacting the buffer pad patterns **131p_1**, **131p_2**, **131p_3** and **131p_4**, respectively. For example, the peripheral contact plugs **286p** may include a first peripheral contact plug **286p_1** penetrating the first insulating region ST_Ia and contacting the first buffer pad pattern **131p_1**, a second peripheral contact plug **286p_2** penetrating the second insulating region ST_Ib and contacting the second buffer pad pattern **131p_2**, and a third peripheral contact plug **286p_3** penetrating the third insulating region ST_Ic and contacting the third buffer pad pattern **131p_3**.

(121) The shape of portions of the first peripheral contact plug **286p_1** and the first contact spacer layer **274s_1** contacting the first buffer pad pattern **131p_1** may be substantially the same as a shape of portions of the first peripheral contact plug **86p_1** and the first contact spacer layer **74s_1** contacting the first peripheral pad pattern **15p_1**. The shape of portions of the second peripheral contact plug **286p_2** and the second contact spacer layer **274s_2** contacting the second buffer pad pattern **131p_2** may be substantially the same as a shape of portions of the second peripheral contact plug **86p_2** and the second contact spacer layer **74s_2** contacting the second peripheral pad pattern **15p_2**, as described with reference to FIG. **4B**. The shape of portions of the third peripheral contact plug **286p_3** and the second contact spacer layer **274s_3** contacting the third buffer pad pattern **131p_3** may be substantially the same as a shape of portions of the third peripheral contact

plug **86p_3** and the third contact spacer layer **74s_3** contacting the third peripheral pad pattern **15p_3**, as described with reference to FIG. 4C.

(122) In some embodiments, the first, second and third peripheral contact plugs **286p_1**, **286p_2**, and **286p_3** and the first, second and third contact spacer layers **274s_1**, **274s_2**, and **274s_3** contacting the first, second and third peripheral contact plugs **286p_1**, **286p_2**, and **286p_3** may be variously modified. Modified examples of the first, second and third peripheral contact plugs **286p_1**, **286p_2** and **286p_3** and the first, second and third contact spacer layers **274s_1**, **274s_2** and **274s_3** will be described in relation to FIGS. 14A, 14B and 14C.

(123) Referring to FIG. 14A, the first, second and third peripheral contact plugs **286p_1**, **286p_2** and **286p_3** and the first, second and third contact spacer layers **274s_1**, **274s_2** and **274s_3** of FIGS. 12A and 12B may be modified in the same manner as the first, second and third peripheral contact plugs **86p_1a**, **86p_2a** and **86p_3a** and the first, second and third contact spacer layers **74s_1a**, **74s_2a**, and **74s_3a** of FIGS. 6A and 6B. Here, FIG. 14A is an enlarged cross-sectional diagram illustrating a portion in which the first peripheral contact plug **286p_1a** and the first contact spacer layer **274s_1a** contacting the first buffer pad pattern **131p_1**.

(124) Referring to FIG. 14B, the first, second and third contact spacer layers **274s_1**, **274s_2**, and **274s_3** in FIGS. 12A and 12B may not be provided, and the first, second and third peripheral contact plugs **286p_1**, **286p_2**, and **286p_3** of FIGS. 12A and 12B may be modified in the same manner as the first, second and third peripheral contact plugs **86p_1a**, **86p_2a**, and **86p_3a** of FIGS. 7A, 7B and 8. Here, FIG. 14 is an enlarged cross-sectional diagram illustrating a portion in which the first peripheral contact plug **286p_1b** contacting the first buffer pad pattern **131p_1**.

(125) Referring to FIG. 14C, the first, second and third peripheral contact plugs **286p_1**, **286p_2**, and **286p_3** and the first, second and third contact spacer layers **274s_1**, **274s_2** and **274s_3** of FIGS. 12A and 12B may be modified in the same manner as the first, second and third peripheral contact plugs **86p_1c**, **86p_2c** and **86p_3c** and the first, second and third contact spacer layers **74s_1b**, **74s_2b** and **74s_3b** illustrated of FIGS. 9A and 9B. Here, FIG. 14C is an enlarged cross-sectional diagram illustrating a portion in which the first peripheral contact plug **286p_1c** contacting the first contact spacer layer **274s_1b**.

(126) An example of a method of manufacturing a semiconductor device according to embodiments of the present disclosure will be described with reference to FIGS. 15A, 15B, 16A, 16B, 16C, 17A and 17B. Here, FIGS. 15A, 16A, and 17A are cross-sectional diagrams taken along line I-I' of FIG. 1, FIGS. 15B, 16B, and 17B are cross-sectional diagrams taken along line II-II' of FIG. 1, and FIG. 16C is a cross-sectional diagram taken along line III-III' of FIG. 1.

(127) Referring to FIGS. 1, 15A, and 15B, the lower structure **3** may include the semiconductor substrate **5**, the peripheral devices **9g** and **10**, the peripheral wiring structure **14**, the pad patterns **15p**, and the lower insulating structure **18** described with reference to FIGS. 1, 2A, 2B and 2C. The lower structure **3** may include a lower pattern layer **22** and an intermediate pattern layer **25** stacked in order on the lower insulating structure **18**, and an upper pattern layer **278** penetrating a portion of the intermediate pattern layer **25** and contacting the lower pattern layer **22**. The lower structure **3** may include the first, second and third gap-fill insulating layers **29a**, **29b**, and **29c** and the outer insulating layer **29d** as described with reference to FIGS. 1, 2A, 2B and 2C.

(128) A lower mold stack structure **ST1** may be formed on the lower structure **3**. The lower mold stack structure **ST1** may include lower interlayer insulating layers **33** and preliminary lower horizontal layers **35** alternately stacked. The lower mold stack structure **ST1** in the staircase region **SA** may be patterned to form a staircase shape, and a lower capping insulating layer **39** covering the staircase portion of the lower mold stack structure **ST1** may be formed.

(129) A sacrificial vertical structure **41** penetrating the lower mold stack structure **ST1** in the memory cell array region **MCA** may be formed.

(130) Referring to FIGS. 1 and 16A, 16B and 16C, an upper mold stack structure **ST2** including upper interlayer insulating layers **43** and preliminary upper horizontal layers **45** alternately stacked

may be formed on the lower mold stack structure ST1. The upper mold stack structure ST2 in the staircase region may be patterned to form a staircase shape, and an upper capping insulating layer 53 covering the staircase shape of the upper mold stack structure ST2 and the lower capping insulating layer 39 may be formed. The capping insulating structure 55 may include the lower and upper capping insulating layers 39 and 53.

(131) In the memory cell array region MCA, a memory vertical structure 58 penetrating the lower and upper mold stack structures ST1 and ST2 may be formed. While the memory vertical structure 58 is formed, the sacrificial vertical structure 41 may be removed. The memory vertical structure 58 may include the channel layer 62, the insulating core pattern 64, the data storage structure 60, and the pad pattern 66 described with reference to FIG. 3A. A first upper insulating layer 69 may be formed on the upper mold stack structure ST2 and the capping insulating structure 55.

(132) Dummy vertical structures 72 penetrating the first upper insulating layer 69, the lower and upper mold stack structures ST1 and ST2, and the capping insulating structure 55 may be formed. Insulating contact pillars 76 may be formed simultaneously with the dummy vertical structures 72. The insulating contact pillars 76 may be formed in positions in which the peripheral contact plugs 86p, the source contact plug 86s, the outer peripheral contact plug 86op, and the contact spacer layers 74s_1, 74s_2, 74s_3, 74s_s, and 74s_o, as described in FIGS. 1, 2A, 2B and 2C.

Accordingly, a portion of the insulating contact pillars 76 may contact the pad patterns 15p, and another portion of the insulating contact pillars 76 may contact the lower pattern layer 22.

(133) The forming the dummy vertical structures 72 and the insulating contact pillars 76 may include simultaneously forming holes penetrating the first upper insulating layer 69, the lower and upper mold stack structures ST1 and ST2, and the capping insulating structure 55 and exposing the lower pattern layer 22 and holes exposing the pad patterns 15p, and forming an insulating material simultaneously filling the holes.

(134) Referring to FIGS. 1, 17A, and 17B, a second upper insulating layer 76 may be formed on the first upper insulating layer 69. Additionally referring to FIGS. 16A, 16B and 16C, trenches penetrating the first and second upper insulating layers 69 and 76, the lower and upper mold stack structures ST1 and ST2, the upper pattern layer 27, and the intermediate pattern layer 25 may be formed, a portion of the trenches formed in the staircase region SA may be formed as a dam structure 80, the intermediate pattern layer exposed by the trenches disposed in the memory cell array region MCA may be substituted for the first intermediate pattern layer 25a described with reference to FIGS. 1, 2A, 2B and 2C. Additionally referring to FIGS. 6A, 6B and 6C, a portion of the preliminary lower and upper horizontal layers 35 and 45 exposed by the remaining trenches disposed in the memory cell array region MCA and the staircase region SA may be substituted for the gate horizontal layers 36g and 46g, a portion of the preliminary lower and upper horizontal layers 35 and 45 disposed in the memory cell array region MCA in the second direction may remain and may be formed as the first insulating region ST_Ia described with reference to FIGS. 1, 2A, 2B and 2C, and the preliminary lower and upper horizontal layers 35 and 45 surrounded by the dam structure 80 in the staircase region SA may be formed as first and second insulating regions ST_Ib and ST_Ic described with reference to FIGS. 1, 2A, 2B and 2C. Accordingly, the stack structure ST as described with reference to FIGS. 1, 2A, 2B and 2C may be formed.

(135) A third upper insulating layer 84 may be formed on the second upper insulating layer 76. Subsequently, the peripheral contact plugs 86p, the source contact plug 86s, the outer peripheral contact plug 86op, and the gate contact plugs 86g as described with reference to FIGS. 1, 2A, 2B and 2C may be simultaneously formed. The forming the peripheral contact plugs 86p, the source contact plug 86s, the outer peripheral contact plug 86op, and the gate contact plugs 86g may include forming peripheral contact holes penetrating the third upper insulating layer 84, extending downwardly, and penetrating the insulating contact pillars 76 and gate contact holes penetrating the first, second and third upper insulating layers 69, 76 and 84 and the capping insulating structure 55, and forming a conductive material simultaneously filling the peripheral contact holes and the gate

contact holes. The insulating contact pillars **76**, the first, second and third upper insulating layers **69**, **76**, and **84**, and the capping insulating structure **55** may be formed of the same material, such as silicon oxide, for example.

(136) In some embodiments, the peripheral contact holes may be formed by etching the insulating contact pillars **76** instead of etching the insulating horizontal layers **36i** and **46i** which may be formed of silicon nitride. Accordingly, since the gate contact holes and the peripheral contact holes are formed by etching the same material, silicon oxide, for example, while the gate contact holes may be formed simultaneously with the peripheral contact holes, a punching defect in which the gate horizontal layers **36g** and **46g** are penetrated by the gate contact holes may be prevented.

(137) Thereafter, a wiring process may be performed to form the bit line **93**, the gate connection wirings **94**, and the source connection wiring **95** described with reference to FIGS. **1**, **2A**, **2B** and **2C**.

(138) In some embodiments, the peripheral contact plugs **86p**, the source contact plug **86s**, the outer peripheral contact plug **86op**, and the gate contact plugs **86g** may be simultaneously formed without defects. Therefore, a height of the semiconductor device may be reduced such that integration density and reliability of the semiconductor device may be improved, and productivity of the semiconductor device may be increased.

(139) FIG. **18** is a conceptual diagram illustrating an electronic system including a semiconductor device according to embodiments of the present disclosure.

(140) Referring to FIG. **18**, an electronic system **1000** may include a semiconductor device **1100** and a controller **1200** electrically connected to the semiconductor device **1100**. The electronic system **1000** may be implemented by a storage device including the semiconductor device **1100** or an electronic device including a storage device. The electronic system **1000** may be a data storage system. For example, the electronic system **1000** may be implemented by a solid state drive device (SSD) device, a universal serial bus (USB), a computing system, a medical device, or a communication device which may include the semiconductor device **1100**.

(141) The semiconductor device **1100** may be implemented by the semiconductor device described in the above-described embodiments with reference to FIGS. **1** to **14C**. The semiconductor device **1100** may include a first structure **110F** and a second structure **1100S** on the first structure **110F**. In embodiments, the first structure **110F** may be disposed neighboring to the second structure **1100S**. The first structure **110F** may be a peripheral circuit structure including a decoder circuit **1110**, a page buffer **1120**, and a logic circuit **1130**. For example, the first structure **110F** may include the peripheral circuit PC, the peripheral wiring structure **14**, and the pad patterns **15p** of the lower structure **3** described above.

(142) The second structure **1100S** may be configured as a memory cell structure including a bit line BL, a common source line CSL, word lines WL, first and second gate upper lines UL1 and UL2, first and second gate lower lines LL1 and LL2, and memory cell strings CSTR disposed between the bit line BL and the common source line CSL.

(143) The pattern structure **20** described above may include a silicon layer having N-type conductivity, and the silicon layer having N-type conductivity may be the common source line CSL.

(144) In the second structure **1100S**, each of the memory cell strings CSTR may include lower transistors LT1 and LT2 adjacent to the common source line CSL, upper transistors UT1 and UT2 adjacent to the bit line BL, and a plurality of memory cell transistors MCT disposed between the lower transistors LT1 and LT2 and the upper transistors UT1 and UT2. The number of the lower transistors LT1 and LT2 and the number of the upper transistors UT1 and UT2 may be varied in embodiments.

(145) In embodiments, the upper transistors UT1 and UT2 may include a string select transistor, and the lower transistors LT1 and LT2 may include a ground select transistor. The lower gate lines LL1 and LL2 may be gate electrodes of lower transistors LT1 and LT2. The word lines WL may be

gate electrodes of memory cell transistors MCT, and the gate upper lines UL1 and UL2 may be gate electrodes of upper transistors UT1 and UT2, respectively.

(146) The gate horizontal layers **36g** and **46g** described above may form the lower gate lines LL1 and LL2, the word lines WL, and the upper gate lines UL1 and UL2. For example, the first and second lower gate horizontal layers **36g_La** and **36g_Lb** as in FIG. 3A may form the lower gate lines LL1 and LL2, and the intermediate gate horizontal layers **36M** and **46M** may form the word lines WL, and the first and second upper gate horizontal layers **46g_Ua** and **46g_Ub** may form the gate upper lines UL1 and UL2.

(147) In embodiments, the lower transistors LT1 and LT2 may include a lower erase control transistor LT1 and a ground select transistor LT2 connected to each other in series. The upper transistors UT1 and UT2 may include a string select transistor UT1 and an upper erase control transistor UT2 connected to each other in series. At least one of the lower erase control transistor LT1 and the upper erase control transistor UT1 may be used to an erase operation for erasing data stored in the memory cell transistors MCT using a gate induced leakage current (GIDL) phenomenon.

(148) The common source line CSL, the first and second gate lower lines LL1 and LL2, word lines WL, and the first and second gate upper lines UL1 and UL2 may be electrically connected to the decoder circuit **1110** through first connection wirings **1115** extending to the second structure **1100S** in the first structure **110F**.

(149) The first connection wirings **1115** may include the gate contact plugs **86g**, the gate connection wirings **94** and the peripheral contact plugs **86p** described above.

(150) The bit lines BL may be electrically connected to the page buffer **1120** through the second connection wirings **1125** extending to the second structure **1100S** in the first structure **110F**.

(151) In the first structure **110F**, the decoder circuit **1110** and the page buffer **1120** may perform a control operation on at least one of the selected memory cell transistors of the plurality of memory cell transistors MCT. The decoder circuit **1110** and the page buffer **1120** may be controlled by a logic circuit **1130**. The semiconductor device **1000** may communicate with the controller **1200** through the input/output pad **1101** electrically connected to the logic circuit **1130**. The input/output pad **1101** may be electrically connected to the logic circuit **1130** through an input/output connection line **1135** extending to the second structure **1100S** in the first structure **110F**.

(152) The controller **1200** may include a processor **1210**, a NAND controller **1220**, and a host interface **1230**. In embodiments, the electronic system **1000** may include a plurality of semiconductor devices **1100**, and in this case, the controller **1200** may control the plurality of semiconductor devices **1100**.

(153) The processor **1210** may control overall operation of the electronic system **1000** including the controller **1200**. The processor **1210** may operate according to a predetermined firmware, and may access the semiconductor device **1100** by controlling the NAND controller **1220**. The NAND controller **1220** may include a NAND interface **1221** for processing communication with the semiconductor device **1100**. Through the NAND interface **1221**, a control command for controlling the semiconductor device **1100**, data to be written in the memory cell transistors MCT of the semiconductor device **1100**, and data to be read from the memory cell transistors MCT of the semiconductor device **1100** may be transmitted. The host interface **1230** may provide a communication function between the electronic system **1000** and an external host. When a control command is received from an external host through the host interface **1230**, the processor **1210** may control the semiconductor device **1100** in response to the control command.

(154) FIG. 19 is a perspective diagram illustrating an electronic system including a semiconductor device according to an embodiment of the present disclosure.

(155) Referring to FIG. 19, an electronic system **2000** may include a main substrate **2001**, a controller **2002** mounted on the main substrate **2001**, one or more semiconductor packages **2003**, and a DRAM **2004**. The semiconductor package **2003** and the DRAM **2004** may be connected to

the controller **2002** by wiring patterns **2005** formed on the main substrate **2001**.

(156) The main substrate **2001** may include a connector **2006** including a plurality of pins coupled to an external host. The number and arrangement of the plurality of pins in the connector **2006** may vary according to a communication interface between the electronic system **2000** and the external host. In embodiments, the electronic system **2000** may communicate with an external host through one of universal serial bus (USB), peripheral component interconnect express (PCI-Express), serial advanced technology attachment (SATA), M-Phy for universal flash storage (UFS), or the like. In embodiments, the electronic system **2000** may operate by power supplied from an external host through the connector **2006**. The electronic system **2000** may further include a power management integrated circuit (PMIC) which may distribute power supplied from the external host to the controller **2002** and the semiconductor package **2003**.

(157) The controller **2002** may write data in the semiconductor package **2003** or may read data from the semiconductor package **2003**, and may improve an operation speed of the electronic system **2000**.

(158) The DRAM **2004** may be configured as a buffer memory for reducing a speed difference between the semiconductor package **2003**, which is a data storage space, and an external host. The DRAM **2004** included in the electronic system **2000** may also operate as a type of cache memory, and may provide a space for temporarily storing data in a control operation on the semiconductor package **2003**. When the DRAM **2004** is included in the electronic system **2000**, the controller **2002** may further include a DRAM controller for controlling the DRAM **2004** in addition to a NAND controller for controlling the semiconductor package **2003**.

(159) The semiconductor package **2003** may include first and second semiconductor packages **2003a** and **2003b** spaced apart from each other. Each of the first and second semiconductor packages **2003a** and **2003b** may be configured as a semiconductor package including a plurality of semiconductor chips **2200**. Each of the semiconductor chips **2200** may include a semiconductor device described in the above-described embodiments with reference to FIGS. **1** to **14C**.

(160) Each of the first and second semiconductor packages **2003a** and **2003b** may include a package substrate **2100**, a semiconductor chips **2200** on the package substrate **2100**, adhesive layers **2300** disposed on a lower surface of each of the semiconductor chips **2200**, a connection structure **2400** electrically connecting the semiconductor chips **2200** to the package substrate **2100**, and a molding layer **2500** covering the semiconductor chips **2200** and the connection structure **2400** on the package substrate **2100**.

(161) The package substrate **2100** may be configured as a printed circuit board including package upper pads **2130**. Each of the semiconductor chips **2200** may include an input/output pad **2210**. The input/output pad **2210** may be the input/output pad **1101** of FIG. **18**.

(162) In embodiments, the connection structure **2400** may be a bonding wire electrically connecting the input/output pad **2210** to the package upper pads **2130**. Accordingly, in each of the first and second semiconductor packages **2003a** and **2003b**, the semiconductor chips **2200** may be electrically connected to each other by a bonding wire method, and may be electrically connected to the package upper pads **2130** of the package substrate **2100**. In embodiments, in each of the first and second semiconductor packages **2003a** and **2003b**, the semiconductor chips **2200** may be electrically connected to each other by a connection structure including a through silicon via (TSV), instead of the connection structure **2400** of a bonding wire method.

(163) In embodiments, the controller **2002** and the semiconductor chips **2200** may be included in a single package. For example, the controller **2002** and the semiconductor chips **2200** may be mounted on an interposer substrate different from the main substrate **2001**, and the controller **2002** and the semiconductor chips **2200** may be connected to the controller **2002** by wiring formed on the interposer substrate.

(164) FIG. **20** is a cross-sectional diagram further illustrating an electronic system including a semiconductor device according to an embodiment of the present disclosure. Here, the

semiconductor package **2003** of FIG. **20** is taken along line IV-IV' in FIG. **19**.

(165) Referring to FIG. **20**, in the semiconductor package **2003**, the package substrate **2100** may be configured as a printed circuit board. The package substrate **2100** may include a package substrate body portion **2120**, package upper pads **2130** disposed on the upper surface of the package substrate body portion **2120**, lower pads **2125** disposed on a lower surface of the package substrate body portion **2120** or exposed through a lower surface, and internal wirings **2135** electrically connecting the package upper pads **2130** to the lower pads **2125** in the package substrate body portion **2120**. The package upper pads **2130** may be electrically connected to the connection structures **2400**. The lower pads **2125** may be connected to the wiring patterns **2005** of the main substrate **2010** of the electronic system **2000** through the conductive connection portions **2800** as illustrated in FIG. **19**.

(166) Each of the semiconductor chips **2200** may include a semiconductor substrate **3010**, and a first structure **3100** and a second structure **3200** stacked in order on the semiconductor substrate **3010**. The first structure **3100** may include a peripheral circuit region including peripheral wirings **3110**. The second structure **3200** may include a common source line **3205**, a gate stack structure **3210** on the common source line **3205**, memory channel structures **3220** and separation structures **3230** penetrating the gate stack structure **3210**, bit lines **3240** electrically connected to the memory channel structures **3220**, and gate connection wirings **94** (e.g., in FIG. **2B**) electrically connected to the word lines WL (e.g., in FIG. **18**) of the gate stack structure **3210**. The first structure **3100** may include the first structure **1100F** of FIG. **18**, and the second structure **3200** may include the second structure illustrated **1100S** of FIG. **18**. For example, in FIG. **20**, an enlarged portion indicated by reference numeral **1** may represent the cross-sectional structure of the example illustrated in FIG. **2B**. Accordingly, each of the semiconductor chips **2200** may include the semiconductor device **1** described in the aforementioned embodiments described with reference to the foregoing drawings.

(167) Each of the semiconductor chips **2200** may include a through wiring **3245** electrically connected to the peripheral wirings **3110** of the first structure **3100** and extending into the second structure **3200**. The through wiring **3245** may penetrate the gate stack structure **3210** and may be further disposed on an external side of the gate stack structure **3210**. The through wiring **3245** may be the peripheral contact plugs **86p** described in one of the aforementioned embodiments described with reference to the foregoing drawings.

(168) Each of the semiconductor chips **2200** may further include an input/output pad **2210** electrically connected to peripheral wirings **3110** of the first structure **3100** and electrically connected to an input/output connection wiring **3265** and an input/output connection wiring **3265** extending into the second structure **3200**.

(169) According to the foregoing embodiments, a semiconductor device (and accordingly, an incorporating electronic device) may have greater integration density and improved reliability.

(170) While the embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A semiconductor device, comprising: a lower structure including a peripheral circuit; a stack structure on the lower structure and including a first stack portion and a second stack portion, wherein the first stack portion includes first interlayer insulating layers and gate conductive layers alternately stacked in a vertical direction, and the second stack portion includes second interlayer insulating layers and third interlayer insulating layers alternately stacked in the vertical direction; a vertical memory structure penetrating through the first stack portion in the vertical direction; a capping insulating layer on the stack structure; a peripheral contact plug penetrating through the capping insulating layer and the second stack portion in the vertical direction; and gate contact

plugs in contact with gate pads of the gate conductive layers, wherein the first interlayer insulating layers and the second interlayer insulating layers include a first material, wherein the third interlayer insulating layers include a second material different from the first material, wherein the peripheral contact plug includes: an upper plug portion penetrating the capping insulating layer and extending downward; a lower plug portion at a lower level than a lowermost third interlayer insulating layer among the third interlayer insulating layers; and an intermediate plug portion between the upper plug portion and the lower plug portion, wherein the intermediate plug portion penetrates the lowermost third interlayer insulating layer and extends upward, wherein the upper plug portion has a first upper side surface and a second upper side surface opposing each other in a first horizontal direction, wherein the intermediate plug portion has a first intermediate side surface and a second intermediate side surface opposing each other in the first horizontal direction, wherein a width of the upper plug portion is greater than a width of the intermediate plug portion, wherein a side surface of the peripheral contact plug includes a bent portion extending from the first intermediate side surface of the intermediate plug portion toward the first upper side surface of the upper plug portion, and wherein at least a portion of the bent portion of the side surface of the peripheral contact plug is at a lower level than an upper surface of an uppermost third interlayer insulating layer among the third interlayer insulating layers.

2. The semiconductor device of claim 1, wherein a first central axis of the upper plug portion and a second central axis of the intermediate plug portion are misaligned.

3. The semiconductor device of claim 2, wherein the second central axis of the intermediate plug portion and a third central axis of the lower plug portion are aligned.

4. The semiconductor device of claim 1, wherein the peripheral contact plug is in contact with the third interlayer insulating layers.

5. The semiconductor device of claim 1, wherein a lower end of the first upper side surface of the upper plug portion is at a lower level than the upper surface of the uppermost third interlayer insulating layer.

6. The semiconductor device of claim 1, wherein a lower end of the first upper side surface of the upper plug portion is at a higher level than a lower surface of the uppermost third interlayer insulating layer.

7. The semiconductor device of claim 1, wherein an upper surface of the peripheral contact plug is at a higher level than an upper surface of the vertical memory structure.

8. The semiconductor device of claim 1, wherein the peripheral contact plug is a plug conductive pattern and a conductive liner layer covering a side surface and a lower surface of the plug conductive pattern.

9. The semiconductor device of claim 1, further comprising: a bit line overlapping with the vertical memory structure and the peripheral contact plug; a first contact plug between the bit line and the vertical memory structure; and a second contact plug between the bit line and the peripheral contact plug.

10. The semiconductor device of claim 1, wherein the vertical memory structure includes: an insulating core pattern; a channel layer on a side surface of the insulating core pattern; and a data storage structure on an external surface of the channel layer.

11. The semiconductor device of claim 1, wherein the first stack portion of the stack structure includes a lower stack structure and an upper stack structure on the lower stack structure, and wherein a side surface of the vertical memory structure includes a bent portion at a level between an uppermost gate conductive layer among the gate conductive layers of the lower stack structure and a lowermost gate conductive layer among the gate conductive layers of the upper stack structure.

12. The semiconductor device of claim 1, wherein the lower structure further includes: a lower insulating structure covering the peripheral circuit; a pattern structure on the lower insulating structure; a pad pattern in the lower insulating structure; and a gap-fill insulating layer penetrating

the pattern structure and overlapping the pad pattern, wherein the pattern structure includes a polysilicon layer, and wherein the vertical memory structure is in contact with the polysilicon layer.

13. The semiconductor device of claim 12, wherein the lower plug portion of the peripheral contact plug extends downwardly from the intermediate plug portion to penetrate the gap-fill insulating layer and contact the pad pattern.

14. The semiconductor device of claim 13, wherein the lower insulating structure includes: a first lower insulating layer at least partially surrounding a side surface of the pad pattern; an etch stop layer on the first lower insulating layer and the pad pattern; and a second lower insulating layer on the etch stop layer, wherein the etch stop layer includes a material different from a material of the second lower insulating layer, wherein a thickness of the etch stop layer is less than a thickness of the second lower insulating layer, and wherein the lower plug portion of the peripheral contact plug penetrates the second lower insulating layer and the etch stop layer to contact the pad pattern.

15. The semiconductor device of claim 14, wherein the lower plug portion of the peripheral contact plug extends further downwardly from the upper surface of the pad pattern into the pad pattern, and a vertical height of the lower plug portion extending from the upper surface of the pad pattern into the pad pattern is greater than a thickness of the etch stop layer.

16. The semiconductor device of claim 1, wherein the lower structure further includes a buffer conductive pattern between a pad pattern and the lower plug portion of the peripheral contact plug.

17. The semiconductor device of claim 16, wherein the vertical memory structure is in contact with a pattern structure, and wherein at least a portion of the buffer conductive pattern is at the same level as at least a portion of the pattern structure.

18. An electronic system, comprising: a main substrate; a semiconductor device on the main substrate; and a controller electrically connected to the semiconductor device, wherein the semiconductor device includes: a lower structure including a peripheral circuit; a stack structure on the lower structure and including a first stack portion and a second stack portion, wherein the first stack portion includes first interlayer insulating layers and gate conductive layers alternately stacked in a vertical direction, and wherein the second stack portion includes second interlayer insulating layers and third interlayer insulating layers alternately stacked in the vertical direction; a vertical memory structure penetrating through the first stack portion in the vertical direction; a capping insulating layer on the stack structure; a peripheral contact plug penetrating through the capping insulating layer and the second stack portion in the vertical direction; and gate contact plugs in contact with gate pads of the gate conductive layers, wherein the first interlayer insulating layers and the second interlayer insulating layers include a first material, wherein the third interlayer insulating layers include a second material different from the first material, wherein the peripheral contact plug includes: an upper plug portion penetrating the capping insulating layer and extending downward; a lower plug portion at a lower level than a lowermost third interlayer insulating layer among the third interlayer insulating layers; and an intermediate plug portion between the upper plug portion and the lower plug portion, wherein the intermediate plug portion penetrates the lowermost third interlayer insulating layer and extends upward, wherein the upper plug portion has a first upper side surface and a second upper side surface opposing each other in a first horizontal direction, wherein the intermediate plug portion has a first intermediate side surface and a second intermediate side surface opposing each other in the first horizontal direction, wherein a width of the upper plug portion is greater than a width of the intermediate plug portion, wherein a side surface of the peripheral contact plug includes a bent portion extending from the first intermediate side surface of the intermediate plug portion toward the first upper side surface of the upper plug portion, and wherein at least a portion of the bent portion of the side surface of the peripheral contact plug is at a lower level than an upper surface of an uppermost third interlayer insulating layer among the third interlayer insulating layers.

19. The electronic system of claim 18, wherein a first central axis of the upper plug portion and a second central axis of the intermediate plug portion are misaligned, and wherein the second central

axis of the intermediate plug portion and a third central axis of the lower plug portion are aligned.

20. The electronic system of claim 18, wherein the peripheral contact plug is in contact with the third interlayer insulating layers, and wherein a lower end of the first upper side surface of the upper plug portion is at a lower level than the upper surface of the uppermost third interlayer insulating layer.
